

PROFILE

AI Engineering student at **Technische Hochschule Ingolstadt** with over 5 years of professional experience as an Embedded Engineer. Specializing in end-to-end **ROS2-based robotic systems** and **Edge AI solutions** including running Vision Language Models on resource-constrained hardware. Expertise bridges complex firmware (C/C++, FreeRTOS) and high-level autonomous behaviors (Python, PyTorch). Having supported **NVIDIA's DRIVE OS** and built autonomous drone solutions at **Hilti**, driven to create production-grade intelligent systems for real-world mobility and logistics.

WORK EXPERIENCE

Robotics & AI Engineer (Work Student)

Hilti | Kaufering, Germany

Dec 2024 - Present

- Architecting an end-to-end **ROS2**-based robotic system for autonomous construction logistics and tool tracking.
- Developed robust localization pipeline using **Particle Filters** for real-time tool tracking in complex environments.
- Integrating **Gaussian Processes** for intelligent path planning and environment modeling to enhance drone navigation efficiency.
- Designing and testing **PX4**- and **ROS2**-based drone control systems with autonomous waypoint navigation and return-to-launch features.
- Integrating computer vision (**LiDAR** + onboard cameras) for obstacle detection and environment awareness.
- Developing and debugging flight behaviors in **Gazebo/SITL** before hardware deployment.
- Supporting research on **Edge AI** applications - lightweight **deep learning** models for real-time decision-making on embedded systems.

Senior Embedded Engineer (Contractor)

NVIDIA via Vayavya Labs | Bengaluru, India

Jul 2023 - Jul 2024

- Supported **DRIVE OS** component of NVIDIA's autonomous driving software stack.
- Collaborated with internal software groups to design and implement the base software layer (**Linux/QNX** OS, device drivers).
- Developed hardware-specific drivers using **AutoSAR** firmware for NVIDIA's upcoming **ADAS** hardware.
- Conducted driver validation through functional test cases; resolved customer-reported **DRIVE OS** bugs.
- Matured prototype software to production-quality standards.

Embedded Systems Software Developer (Contract)

Stasis Labs | Bengaluru, India

Apr 2023 - Jul 2023

- Decoded **medical device** data using manufacturer communication protocol specifications.
- Designed and developed application software to read, format, and transmit data from **medical devices**.
- Collaborated with hardware team on cable design and **signal-level validation** for reliable data transmission.
- Wrote automation scripts for task scheduling, device simulators, and test rigs.

Embedded Engineer

Frenustech | Bengaluru, India

Aug 2021 - Oct 2022

- Developed FPGABSP for a custom **RISC-V** core; implemented and validated **I2C, UART, SPI** drivers on XilinxARTYA7.
- Implemented real-time **object detection** on STM32H747I (dual-core ARM) using a compressed **TensorFlow Lite** model.
- Developed **FreeRTOS**-based applications and **multi-sensor fusion** firmware for STM32 microcontrollers.
- Designed SDK for ARM-based microcontrollers; conducted performance benchmarking.

UAV Researcher Intern

Vyomastra | Bangalore, India

Mar 2021 - Sep 2021

- Developed **UAV firmware** for photogrammetry and AI-based drone applications.
- Designed UAV subsystems: **battery management, flight controller**, and **NPNT** compliance modules.

TECHNICAL SKILLS

Robotics & Autonomy	AI & Edge Inference	Embedded & Systems
ROS2, PX4, Gazebo, NVIDIA Isaac Sim, NVIDIA Omniverse, Isaac ROS, OpenCV, SLAM, Particle Filters, Path Planning	PyTorch, TensorFlow Lite, ONNX, TensorRT, INT8 / FP16 Quantization, Model Pruning, DeepStream, Edge AI, VLMs	C / C++, Embedded C, AutoSAR, FreeRTOS, Python, Linux, QNX, STM32, Git, Jupyter

EDUCATION

M.Eng. - AI Engineering of Autonomous Systems Technische Hochschule Ingolstadt Germany Thesis: Enhancing UAV Control with Large Language Models: Natural Language Interfaces and Context-Aware Guidance	Oct 2024 - Present
B.Eng. - Electronics and Communication Engineering Vidyavardhaka College of Engineering Mysuru, India Final Grade: 1.7	Aug 2017 - Aug 2021

PROJECTS

Autonomous Object Sorting & Transport - Dual-Robot Collaboration • DJI RoboMaster EP + Arduino Braccio arm for warehouse-style object sorting and transportation. • ArUco marker + OpenCV navigation - 85% alignment accuracy . • HSV color detection: 95% (red) and 90% (green) under varying lighting. • Multi-threaded control across Raspberry Pi and Windows PC with real-time sync.	Apr 2025 - Jul 2025
Custom UAV Flight Controller & Transceiver Stack Final Year Project • Designed full UAV flight controller firmware from ground up: sensor fusion, control loops, and RF comms. • IMU sensor fusion (accel, gyro, magnetometer) for real-time attitude estimation. • PID attitude stabilization and altitude hold tuned through iterative flight tests. • Custom RF transceiver for real-time telemetry and remote command reception.	Oct 2020 - Apr 2021
GSM-Based Real-Time Location Tracking System • GPS + GSM real-time asset tracking with cellular SMS-based location transmission. • Microcontroller firmware to parse NMEA sentences and format data for reliable output.	Jan 2019 - Mar 2019

LANGUAGE SKILLS

English	C1, Proficient
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HONOURS & AWARDS

National Level Hackathon Winner ATME Engineering College	Mar 2019
National Level Hackathon Winner PES College, Mandya	Oct 2019